

① BCS Driven system:

$$\dot{p} = -i[H_0, p] + \gamma_0 D[p]$$

$$H^{MC}(t) = \sum_k (c_k^\dagger, c_{-k}) \begin{pmatrix} \xi_k & -\Delta \\ -\Delta & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \xi_k$$

$$= (c_k^\dagger, c_{-k}) \begin{pmatrix} \xi_k c_k + \Delta c_{-k}^\dagger \\ \Delta c_k - \xi_k c_{-k}^\dagger \end{pmatrix} =$$

$$= c_k^\dagger c_k \xi_k + \Delta c_k^\dagger c_{-k}^\dagger + \Delta c_{-k} c_k - \xi_k c_k c_{-k}^\dagger =$$

$$= \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) + \Delta (c_k^\dagger c_{-k}^\dagger + c_{-k} c_k) - \xi_k c_k c_{-k}^\dagger$$

Bozonsymmetrie: $\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} m & r \\ -r & m \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}$

↳ gleich Netto: x opp

gap: $\Delta_k(t) = \sum_{nn'} V_{nn'} \text{Tr}[c_k^\dagger c_{k'} p(t)]$

$$\rightarrow E_k(t) = \pm \sqrt{\xi_k^2 + \Delta_k^2(t)} \quad \text{in} \quad N_k(t) = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k(t)} \right), \quad N_{-k} = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k(t)} \right)$$

→ simple modellierung kopeln:

$$D[\cdot] = D^{(T_1)}[\cdot] + D^{(T_2)}[\cdot]$$

$$D^{(T_1)} p = \sum_{\substack{\text{r.f.} \\ k}} L_p L^\dagger - \frac{1}{2} \{p, L L^\dagger\}$$

$$L_{kp}^{(s,r)} = \sqrt{\gamma(t)} \alpha_k^{(s)} \alpha_p^{(r)}$$

$$\alpha^{(+)} = \alpha^\dagger$$

$$\alpha^{(-)} = \alpha$$

$$k = (k, \sigma)$$

$$p(t) = \frac{1}{2} \exp \left[\sum_k \mu_k(t) \alpha_k^\dagger \alpha_k \right]$$

$$\langle n_k \rangle = n_k = \langle \alpha_k^\dagger \alpha_k \rangle$$

trace

$$\rightarrow \text{EOM: } \langle \dot{n}_k \rangle = \text{Tr}[n_k D p] = \gamma_0 \sum_j \left(\langle L_j^\dagger n_k L_j \rangle - \langle L_j^\dagger L_j n_k \rangle \right)$$

$$s_1 s_2 = \pm 1$$

$$\rightarrow \langle L_j^\dagger n_k L_j \rangle \propto \sum_{p_2} \langle \alpha_p^\dagger \alpha_{-p}^\dagger \alpha_{-k}^\dagger \alpha_k^\dagger \alpha_{-p} \alpha_p \rangle \stackrel{!}{=} \text{6 messung.}$$

$$\langle \alpha_k^\dagger \alpha_p \rangle = n_k \delta_{ks}$$

EOM:

$$\langle \dot{n}_k \rangle = \text{Tr}[n_k(t) \dot{p}(t)] \stackrel{!}{=} -i \text{Tr}[n_k(t) [H(t), p(t)]] + \gamma_0 \text{Tr}[n_k(t) D p] =$$

$$= \gamma_0 \text{Tr}[n_k (D_1 p + D_2 p)] = \gamma_0 \text{Tr}[n_k D^{(T_1)} p] \quad \text{mit } D_1 = D^{(T_1)}, \quad L_1 = \sqrt{\gamma(t)} \alpha_k^\dagger \alpha_{-k}^\dagger \dots$$

$$= r_0 \text{Tr} \left(n_k \sum_{\substack{s_1 s_2 \\ 2p}} \gamma_{2p}^{s_1 s_2} (T_1) \alpha_p^{s_1} \alpha_p^{s_2} \left\{ \alpha_p^{s_1+} \alpha_p^{s_2+} - \frac{1}{2} \delta_{2p} \right\} \rho, \alpha_p^{s_1+} \alpha_p^{s_2+} \alpha_p^{s_1} \alpha_p^{s_2} \right) + r_0 (T_2) =$$

$$= r_0 \sum_{\substack{s_1 s_2 \\ 2p}} \gamma_{2p}^{s_1 s_2} (T_1) \left(\langle \alpha_p^{s_1+} \alpha_p^{s_2+} n_k \alpha_p^{s_1} \alpha_p^{s_2} \rangle - \langle \alpha_p^{s_1+} \alpha_p^{s_2+} \alpha_p^{s_1} \alpha_p^{s_2} n_k \rangle \right) + r_0 (T_2) =$$

$$\text{Tr} [f \frac{1}{2} \{p, A\}] = \text{Tr} \left[\frac{1}{2} f p A + \frac{1}{2} f A p \right] = \frac{1}{2} \text{Tr} [f p A + f A p]$$

$$= \frac{1}{2} \text{Tr} [(A f + f A) \cdot p] = \frac{1}{2} \langle A f + f A \rangle$$

if $f = n_k \rightarrow \langle A f \rangle$

→ compare: $s_1 s_2 = +1$

$$\textcircled{++} \langle \alpha_p \alpha_z \alpha_k^+ \alpha_k^+ \alpha_z^+ \alpha_p^+ \rangle - \langle \alpha_p \alpha_z \alpha_z^+ \alpha_p^+ \alpha_k^+ \alpha_k^+ \rangle = \langle [n_k, \alpha_p^+] \rangle = \langle \alpha_k^+ \alpha_k, \alpha_p^+ \rangle =$$

$$= \langle \alpha_p \alpha_z [n_k, \alpha_z^+ \alpha_p^+] \rangle =$$

$$[n_k, \alpha_z^+ \alpha_p^+] = [n_k, \alpha_z^+] \alpha_p^+ + \alpha_z^+ [n_k, \alpha_p^+] = \alpha_k^+ \alpha_p^+ \delta_{kz} + \alpha_z^+ \alpha_k^+ \delta_{kp}$$

$$= \langle \alpha_p \alpha_z (\alpha_k^+ \alpha_p^+ \delta_{kz} + \alpha_z^+ \alpha_k^+ \delta_{kp}) \rangle = (\delta_{kz} + \delta_{kp}) \langle \alpha_p \alpha_z \alpha_z^+ \alpha_p^+ \rangle =$$

$$\alpha \alpha^+ = 1 - \alpha^+ \alpha = 1 - n$$

$$= (\delta_{kz} + \delta_{kp}) \cdot (1 - \delta_{2p}) \langle \alpha_p \alpha_p^+ \alpha_z \alpha_z^+ \rangle =$$

$$= (\delta_{kz} + \delta_{kp}) (1 - \delta_{2p}) (1 - n_p) (1 - n_z)$$

$$\boxed{T_{2p} = T_{12}}$$

$$\dot{n}_k = r_0 \sum_{2p} \Gamma_{2p}^{++} (\delta_{kz} + \delta_{kp}) (1 - \delta_{2p}) (1 - n_p) (1 - n_z) + 3 \text{ term} =$$

$$= r_0 \left(\sum_p \left[\Gamma_{kp}^{++} (1 - \delta_{kp}) (1 - n_p) (1 - n_k) \right] + \sum_{\substack{z \\ 2 \rightarrow p}} \left[\Gamma_{kz}^{++} (1 - \delta_{kz}) (1 - n_k) (1 - n_z) \right] \right) =$$

$$= r_0 \cdot 2 \sum_p \Gamma_{kp}^{++} (1 - \delta_{kp}) (1 - n_p) (1 - n_k) + 3 \text{ term}$$

$$2 \sum \Gamma (1-n)(1-n) = r_0 2 \Gamma_{kk}^{++} (1-n_k)^2 = \dots = 2 r_0 \Gamma_{kk}^{++} n_k$$

prova δ_1 ? 0.1 ? ?

problem is $(--)$: $\langle \alpha_p^\dagger \alpha_z^\dagger n_k \alpha_z \alpha_p \rangle - \langle \alpha_p^\dagger \alpha_z^\dagger \alpha_z \alpha_p n_k \rangle =$

$\begin{pmatrix} s_1 = - \\ s_2 = - \end{pmatrix} \rightarrow \langle \alpha_p^\dagger \alpha_z^\dagger [n_k, \alpha_z \alpha_p] \rangle = \langle [n_k, \alpha_p] \alpha_z^\dagger \alpha_p \rangle = \langle \alpha_k^\dagger \alpha_p \rangle - \delta_{kp} \alpha_k$

$$[n_k, \alpha_p \alpha_z] = \alpha_p [n_k, \alpha_z] + [n_k, \alpha_p] \alpha_z =$$

$$= -(\alpha_p \alpha_k \delta_{kz} + \alpha_k \alpha_z \delta_{kp})$$

$$= -\langle \alpha_p^\dagger \alpha_z^\dagger (\alpha_p \alpha_k \delta_{kz} + \alpha_k \alpha_z \delta_{kp}) \rangle = -\langle \alpha_p^\dagger \alpha_z^\dagger \alpha_p \alpha_z \rangle (\delta_{kz} + \delta_{kp}) = -n_z n_p (1 - \delta_{zp}) (\delta_{kz} + \delta_{zp})$$

-1 2. term: $-\gamma_0 \left(\sum_p \Gamma_{kp}^{--} n_k n_p (1 - \delta_{kp}) + \sum_z \Gamma_{kz}^{--} n_z n_k (1 - \delta_{kz}) \right) =$

$$= -2\gamma_0 \sum_p \Gamma_{kp}^{--} n_p n_k - 2\gamma_0 n_k^2 \Gamma_{kk}^{--} n_k \leftarrow \text{problem with } \Gamma_k?$$

$(+-)$ $\begin{pmatrix} s_1 = + \\ s_2 = - \end{pmatrix}$

$$\langle \alpha_p^\dagger \alpha_z^\dagger n_k \alpha_z^\dagger \alpha_p \rangle - \langle \alpha_p^\dagger \alpha_z^\dagger \alpha_z^\dagger \alpha_p n_k \rangle =$$

$$= \langle \alpha_p^\dagger \alpha_z^\dagger [n_k, \alpha_z^\dagger \alpha_p] \rangle = \langle \alpha_p^\dagger \alpha_z^\dagger (\alpha_k^\dagger \alpha_p \delta_{kz} - \alpha_z^\dagger \alpha_k \delta_{kp}) \rangle =$$

$$= [n_k, \alpha_z^\dagger] \alpha_p^\dagger + \alpha_z^\dagger [n_k, \alpha_p] = \alpha_k^\dagger \alpha_p \delta_{kz} - \alpha_z^\dagger \alpha_k \delta_{kp}$$

$$= \langle \alpha_p^\dagger \alpha_z^\dagger \alpha_k^\dagger \alpha_p \rangle (\delta_{kz} - \delta_{kp}) = (\delta_{kz} - \delta_{kp}) n_p (1 - n_z)$$

$$\langle \alpha_p^\dagger \alpha_z^\dagger \alpha_z^\dagger \alpha_p \rangle (-\delta_{kz} + \delta_{kp}) = -(\delta_{kz} - \delta_{kp}) (1 - n_p) n_z$$

$\rightarrow$ Lemma: $\Gamma_{pz}^{+-} \neq \Gamma_{zp}^{+-}!$

$$\gamma_0 \sum_{p,z} \Gamma_{pz}^{+-} (\delta_{kz} - \delta_{kp}) (n_p (1 - n_z) - n_z (1 - n_p)) =$$

$$= \gamma_0 \left(\sum_p \Gamma_{kp}^{+-} (n_p (1 - n_k) - n_k (1 - n_p)) - \sum_{p,z} \Gamma_{pz}^{+-} (n_z (1 - n_p) - n_p (1 - n_z)) \right) =$$

$$= \gamma_0 \left(\sum_p \Gamma_{kp}^{+-} (n_p - n_k) - \Gamma_{pk}^{+-} (n_k - n_p) \right) = \{ \text{what?} \}$$

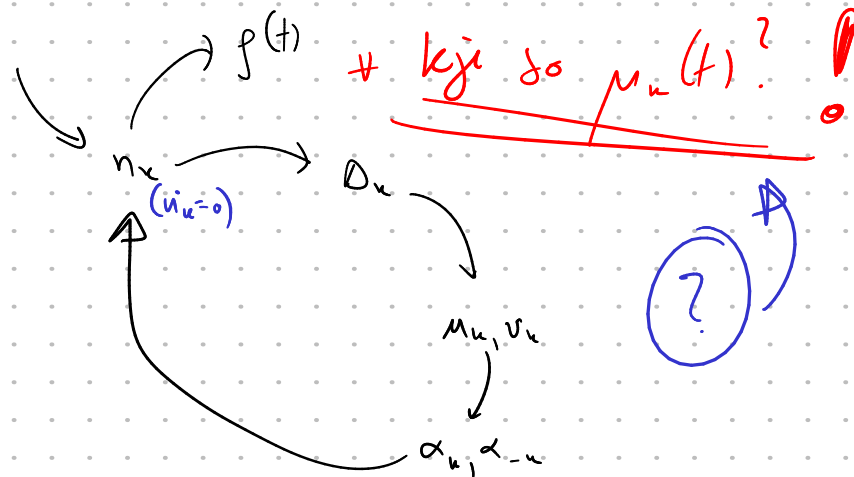
degenerate?

$$\left\{ \begin{aligned} & \sum_{p,z} \Gamma_{pz}^{+-} (\delta_{kz} - \delta_{kp}) n_p (1 - n_z) - \sum_{p,z} \Gamma_{zp}^{+-} (\delta_{kz} - \delta_{kp}) n_z (1 - n_p) = \sum_{p,z} (\Gamma_{pz}^{+-} - \Gamma_{zp}^{+-}) (\delta_{kz} - \delta_{kp}) n_p (1 - n_z) \\ & = \sum_{p,z} \tilde{\Gamma}_{pz} (\delta_{kz} - \delta_{kp}) n_p (1 - n_z) = \left(\sum_p \tilde{\Gamma}_{kp} n_p (1 - n_k) - \sum_p \tilde{\Gamma}_{pk} n_k (1 - n_p) \right) \end{aligned} \right.$$

$\leftarrow$ as before, $\tilde{\Gamma} \neq \Gamma^{+-}$

skupaj: (1) enota za $\dot{n}_k(t) = f(n_k, \dots)$

(2) gap: $\Delta_k(t) = \sum_{k'} \frac{V_{kk'}}{2} \frac{1}{\sqrt{\xi_{k'}^2 + \Delta_{k'}^2(t)}} \cdot (1 - n_k(t) - n_{-k}(t))$



skupaj:

(+)

$$\sum_{2p} \Gamma_{2p}^{+-} (\delta_{k2} - \delta_{kp}) n_p (1 - n_2) = \sum_p (\Gamma_{kp}^{+-} n_p (1 - n_k) - \Gamma_{p2}^{+-} n_k (1 - n_p)) \quad (+)$$

(-+) $\Gamma_{2p}^{--} = \Gamma_{p2}^{++}$ (?) preveri!

$$-\sum_{2p} \Gamma_{2p}^{--} (\delta_{k2} - \delta_{kp}) n_2 (1 - n_p) = \sum_p \Gamma_{pk}^{+-} (-n_k (1 - n_p)) + \sum_p \Gamma_{kp}^{+-} n_p (1 - n_k) = \sum_p (\Gamma_{kp}^{+-} n_p (1 - n_k) - \Gamma_{pk}^{+-} n_k (1 - n_p)) \quad (-)$$

samo en
člen!

$$\dot{n}_k = 0 = 2 \sum_j \left(\Gamma_{kp}^{++} (1-n_k)(1-n_p) - \Gamma_{kp}^{--} n_k n_p \right) + \sum_p \left(\Gamma_{kp}^{+-} (n_p - n_k) - \Gamma_{pk}^{+} (n_k - n_p) \right) = \quad ?$$

$$\alpha = \frac{\alpha_{++}}{\alpha_{++} + \alpha_{--}} \quad , \quad \gamma_{s_1 s_2}^{s_1 s_2} = \alpha_{s_1 s_2} f_{s_1} \left(\frac{\epsilon_k}{T} \right) f_{s_2} \left(\frac{\epsilon_p}{T} \right)$$

$$\begin{aligned} \rightarrow & \quad 2 \alpha_{++} \left(f_{+} \left(\frac{\epsilon_k}{T_1} \right) f_{+} \left(\frac{\epsilon_p}{T_1} \right) + f_{+} \left(\frac{\epsilon_k}{T_2} \right) f_{+} \left(\frac{\epsilon_p}{T_2} \right) \right) (1-n_k)(1-n_p) - \\ & - 2 \alpha_{--} \left(f_{-} \left(\frac{\epsilon_k}{T_1} \right) f_{-} \left(\frac{\epsilon_p}{T_1} \right) + f_{-} \left(\frac{\epsilon_k}{T_2} \right) f_{-} \left(\frac{\epsilon_p}{T_2} \right) \right) n_k n_p + \\ & + \alpha_{+-} \left[\left(f_{+} \left(\frac{\epsilon_k}{T_1} \right) f_{-} \left(\frac{\epsilon_p}{T_1} \right) + f_{-} \left(\frac{\epsilon_k}{T_2} \right) f_{+} \left(\frac{\epsilon_p}{T_2} \right) \right) (1-n_k)n_p - \right. \\ & \left. - \left(f_{+} \left(\frac{\epsilon_p}{T_1} \right) f_{-} \left(\frac{\epsilon_k}{T_1} \right) + f_{+} \left(\frac{\epsilon_p}{T_2} \right) f_{-} \left(\frac{\epsilon_k}{T_2} \right) \right) n_k(1-n_p) \right] = \quad \text{resumo} \quad \text{(kind of)} \quad \text{???} \end{aligned}$$

$$\begin{aligned} = & \sum_{T_j} \left[\sum_p \left(f_{+} \left(\frac{\epsilon_p}{T_j} \right) (1-n_p) + \alpha f_{-} \left(\frac{\epsilon_p}{T_j} \right) n_p \right) \cdot (1-n_k) f_{+} \left(\frac{\epsilon_k}{T_j} \right) - \right. \\ & \left. - \left(\sum_p \left(f_{-} \left(\frac{\epsilon_p}{T_j} \right) n_p + \alpha f_{+} \left(\frac{\epsilon_p}{T_j} \right) (1-n_p) \right) n_k f_{-} \left(\frac{\epsilon_k}{T_j} \right) \right] = \dot{n}_k = \end{aligned}$$

$$= \sum_{T_j} \left(A(T_j) (1-n_k) f_{+} \left(\frac{\epsilon_k}{T_j} \right) - B(T_j) n_k f_{-} \left(\frac{\epsilon_k}{T_j} \right) \right) = \dot{n}_k = 0$$

Fokscitandni i tobatangi + thermal baths:

$$H = \sum_k (a_k^\dagger b_k^\dagger) \begin{pmatrix} \epsilon_a + V_{ab}(t) & \Delta_k^*(t) \\ \Delta_k(t) & \epsilon_b + V_{ba}(t) \end{pmatrix} \begin{pmatrix} a_k(t) \\ b_k(t) \end{pmatrix} \rightarrow \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} = \begin{pmatrix} u & v \\ -v^* & u \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}$$

$$\Delta_k(t) = -\sum_{k'} V_{kk'} \text{Tr}[a_{k'}^\dagger b_{k'} \rho(t)]$$

kann tu raten was μ , da je $n_a + n_b = 1$!

$$E_{\pm}(t) = \frac{\epsilon_a(t) + \epsilon_b(t)}{2} \pm \sqrt{\left(\frac{\epsilon_a - \epsilon_b}{2}\right)^2 + \Delta_k^2} = \eta_k(t) \pm \sqrt{\xi_k^2(t) + \Delta_k^2(t)}$$

$$\Delta_k(t) = -\sum_{k'} \frac{V_{kk'}}{2} \cdot \frac{1}{\sqrt{\xi_{k'}^2(t) + \Delta_{k'}^2(t)}} \cdot (\langle \alpha_k^\dagger \alpha_{k'} \rangle - \langle \beta_k^\dagger \beta_{k'} \rangle)$$

$$u_k = \frac{1}{2} \left(1 + \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta_k^2}} \right), \quad v_k = \frac{1}{2} \left(1 - \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta_k^2}} \right)$$

$$\vec{d}_k = \begin{pmatrix} d_{\alpha k} \\ d_{\beta k} \end{pmatrix} = \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}, \quad \vec{n}_k = \begin{pmatrix} n_{\alpha k} \\ n_{\beta k} \end{pmatrix} = \begin{pmatrix} \langle \alpha_k^\dagger \alpha_k \rangle \\ \langle \beta_k^\dagger \beta_k \rangle \end{pmatrix}$$

$$\bullet \text{ 2. } L_{2p}^{\lambda\lambda'} = \sqrt{\gamma_{2p}^{\lambda\lambda'}} d_{\lambda 2}^\dagger d_{\lambda' p} \quad \left\{ \text{same number preserving? } \lambda\lambda' = \{(\alpha\alpha), (\beta\beta), (\alpha\beta), (\beta\alpha)\} \right\}$$

$$\Rightarrow \dot{n}_{\alpha k} = \text{Tr}[n_{\alpha k} \mathcal{Q}(t)] = \sum_{\substack{2p \\ \lambda\lambda'}} \text{Tr} \left[n_{\alpha k} \left(L_{2p}^{\lambda\lambda'} \rho(t) L_{2p}^{\lambda\lambda'+\dagger} - \frac{1}{2} \{ L_{2p}^{\lambda\lambda'+\dagger} L_{2p}^{\lambda\lambda'}, \rho \} \right) \right] =$$

$$= \sum_{\substack{2p \\ \lambda\lambda'}} \langle L^{\dagger}, [n, L] \rangle = ?$$

$$[n_{\alpha k}, d_{\lambda 2}^\dagger d_{\lambda' p}] = d_{\lambda 2}^\dagger [n_{\alpha k}, d_{\lambda' p}] + [n_{\alpha k}, d_{\lambda 2}^\dagger] d_{\lambda' p} = \textcircled{0}$$

$$\text{denn } \{d_{\alpha k}, d_{\lambda' p}\} = \{d_{\alpha k}^\dagger, d_{\lambda' p}\} = 0 \Rightarrow -d_{\alpha k}^\dagger \delta_{\lambda\lambda'} d_{\lambda' p}$$

$$\textcircled{0} = -d_{\lambda 2}^\dagger d_{\lambda' p} \delta_{\lambda\lambda'} \delta_{\alpha\lambda} + d_{\lambda 2}^\dagger d_{\lambda' p} \delta_{\alpha\lambda} \delta_{\lambda\lambda'} =$$

$$= \sum_{\substack{2p \\ \lambda\lambda' \\ n_1, n_2}} \left[\langle d_{\lambda' p}^\dagger d_{\lambda 2} \rangle \gamma_{2p}^{\lambda\lambda'} \delta_{\alpha\lambda} \delta_{\lambda\lambda'} + \langle d_{\lambda' p}^\dagger d_{\lambda 2} d_{\lambda 2}^\dagger d_{\lambda' p} \rangle \delta_{\alpha\lambda} \delta_{\lambda\lambda'} \gamma_{2p}^{\lambda\lambda'} \right] =$$

$$= \sum_{p, \lambda'} \Gamma_{kp}^{\lambda\lambda'} n_{\lambda' p} (1 - n_{\alpha k}) - \sum_{2p} \Gamma_{2p}^{\lambda\lambda} n_{\alpha k} (1 - n_{\lambda p}) =$$

$$\dot{n}_{\alpha k} = \sum_{\substack{p, \lambda \\ \lambda' = \{\alpha, \beta\}}} \left[\Gamma_{kp}^{\lambda\lambda'} n_{\lambda' p} (1 - n_{\alpha k}) - \Gamma_{2p}^{\lambda\lambda} n_{\alpha k} (1 - n_{\lambda p}) \right]$$

↓ $w = \alpha$ fixed: cual?

$$\dot{V}_{\alpha u} = \sum_i \left[\begin{aligned} & \tau_{kp}^{\alpha\alpha} n_{ap} (1 - n_{ak}) - \\ & - \tau_{pk}^{\alpha\alpha} n_{ak} (1 - n_{ap}) + \\ & + \tau_{kp}^{\alpha\beta} n_{bp} (1 - n_{ak}) - \\ & - \tau_{pk}^{\beta\alpha} n_{ak} (1 - n_{bp}) \end{aligned} \right]$$

delec α
delec β

(p, p, p, p, p, p)

- ① α iz $p \rightarrow k$ (ic uizondan)
② α iz $k \rightarrow p$ (ic k β)

→ detailed balance:

① $\gamma_{kp}^{\alpha\alpha} = e^{-\frac{(\epsilon_k + \epsilon_p)}{\tau}} \gamma_{pk}^{\beta\beta}$?

② $\gamma_{kp}^{\alpha\beta} e^{\frac{\epsilon_k - \epsilon_p}{\tau}} = \gamma_{pk}^{\alpha\beta}$

→ $\gamma_{kp}^{\alpha\alpha}(\tau) = \alpha_{kp} f_p\left(\frac{\epsilon_k}{\tau}\right) f_k\left(\frac{\epsilon_p}{\tau}\right) \rightarrow f_{\alpha} = \frac{1}{1 + e^{\pm \epsilon_k/\tau}}$?

$\alpha_{ap} = \alpha_{pp}$ $\alpha = \frac{d_{ap}}{d_{aa}}$

→ SS: $0 = \dot{V}_{\alpha u} = \sum_{p,j} \left(\alpha_{ap} f_a\left(\frac{\epsilon_p}{\tau_j}\right) f_a\left(\frac{\epsilon_j}{\tau_j}\right) [n_{ap} (1 - n_{ak}) - n_{ak} (1 - n_{ap})] \right.$

$+ \alpha_{ap} f_a\left(\frac{\epsilon_p}{\tau_j}\right) f_p\left(\frac{\epsilon_p}{\tau_j}\right) n_{bp} (1 - n_{ak}) -$

$\left. - \alpha_{ap} f_p\left(\frac{\epsilon_p}{\tau_j}\right) f_a\left(\frac{\epsilon_p}{\tau_j}\right) n_{ak} (1 - n_{bp}) \right) =$

$= \sum_j \left(f_a\left(\frac{\epsilon_j}{\tau_j}\right) \cdot ? - f_p\left(\frac{\epsilon_j}{\tau_j}\right) \cdot ? \right)$

~~$\left(\sum_p f_a\left(\frac{\epsilon_p}{\tau_j}\right) (1 - n_{ap}) + \alpha (1 - n_{bp}) n_{ak} \right)$~~

$\underbrace{\left(\sum_p f_a\left(\frac{\epsilon_p}{\tau_j}\right) n_{ap} + \alpha f_p\left(\frac{\epsilon_j}{\tau_j}\right) n_{bp} \right)}_{A(\tau_j)} (1 - n_{ak})$

$= \sum_j \left(f_a\left(\frac{\epsilon_j}{\tau_j}\right) \left\{ \sum_p f_a\left(\frac{\epsilon_p}{\tau_j}\right) n_{ap} + \alpha f_p\left(\frac{\epsilon_j}{\tau_j}\right) n_{bp} \right\} (1 - n_{ak}) - f_p\left(\frac{\epsilon_j}{\tau_j}\right) \left\{ \sum_p f_a\left(\frac{\epsilon_p}{\tau_j}\right) (1 - n_{ap}) + \alpha (1 - n_{bp}) n_{ak} \right\} n_{ak} \right) =$

$= \sum_{ij} \left(n_{ap} (1 - n_{ak}) f_a^i f_a^j + \alpha n_{bp} (1 - n_{ak}) f_a^i f_p^j - n_{ak} (1 - n_{ap}) f_p^i f_a^j - \alpha n_{ak} (1 - n_{bp}) f_p^i f_p^j \right)$

$$\begin{aligned}
 &= \sum_i \left((1-n_{ak}) f_a^k \right) \left\{ n_{ab} f_a^b \right\} + (1-n_{ak}) f_a^k \left\{ \alpha n_{ab} f_a^b \right\} - n_{ak} f_a^k \left\{ (1-n_{ap}) f_a^p \right\} - n_{ak} f_a^k \left\{ \alpha (1-n_{ap}) f_a^p \right\} \\
 &= \sum_j \left(f_a \left(\frac{c_k}{T_j} \right) (1-n_{ak}) \left[\sum_i f_a \left(\frac{c_i}{T_j} \right) n_{ap} + \alpha n_{ab} f_a \left(\frac{c_i}{T_j} \right) \right] - f_a \left(\frac{c_k}{T_j} \right) n_{ak} \left[\sum_i (1-n_{ap}) f_a \left(\frac{c_i}{T_j} \right) \right] - \right. \\
 &\quad \left. - f_a \left(\frac{c_k}{T_j} \right) n_{ak} \left[\alpha (1-n_{ap}) f_a \left(\frac{c_i}{T_j} \right) \right] \right) =
 \end{aligned}$$

$$\dot{n}_a = \sum_j \left(f_a \left(\frac{c_k}{T_j} \right) \left\{ (1-n_{ak}) A - n_{ak} B \right\} + f_a \left(\frac{c_k}{T_j} \right) n_{ak} C \right) = 0$$

$$\dot{n}_b = \sum_j \left(f_b \left(\frac{c_k}{T_j} \right) \left\{ (1-n_{bk}) D - n_{bk} E \right\} + f_b \left(\frac{c_k}{T_j} \right) n_{bk} F \right) = 0$$

$$n_{pk} = \frac{- \sum_j f_a^k (1-n_{ak}) A - n_{ak} f_a^k B}{\sum_j f_a^k C}$$

$$\begin{aligned}
 0 &= \cancel{\sum_j f_p^k} + \sum_j f_a^k ((1-n_{ak}) A - n_{ak} B) \left(\frac{D}{C} + \frac{E}{C} \right) + \sum_j f_a^k n_{ak} F = 0 - \sum_j f_p^k \\
 &\quad (1-n_{ak}) \sum_j f_a^k A + n_{ak} \sum_j f_a^k B \quad n_{ak} \sum_j f_a^k F = - \sum_j f_p^k - \sum_j f_a^k A \left(\frac{D}{C} + \frac{E}{C} \right) \\
 n_{ak} &\left(- \sum_j f_a^k \left(\left(\frac{D}{C} + \frac{E}{C} \right) (A+B) \right) - F \right) =
 \end{aligned}$$

$$n_{ak} = \frac{\sum_j f_a^k + f_a^k \frac{A}{C} (D+E)}{\sum_j f_a^k \left[\frac{(A+B)}{C} (D+E) - F \right]}$$

2

неприменно

iterativni loop: (1.) $n_a - n_b = m = \sum_k (\mu_k^L - \nu_k^L) (n_{ak} - n_{bk})$

(2) $\Delta = \sum_k \frac{1}{N_k} V \cdot \mu_k \nu_k (n_{bk} - n_{ak}) = \sum_k \frac{V}{N_k} \frac{\Delta}{2\sqrt{\xi_k^2 + \Delta^2}} (n_{bk} - n_{ak})$

(3) $n_o = 1 = n_a + n_b = \sum_k (n_{ak} + n_{bk})$ — bij dele to? (doleto npru?)

(4) $I_{ak} = 0 = \dot{n}_{ak} = \sum_p \left(\Gamma_{kp}^{\alpha\alpha} (n_{kp} - n_{ak}) + \Gamma_{kp}^{\alpha\beta} n_{kp} (1 - n_{ak}) - \Gamma_{pk}^{\beta\alpha} n_{ak} (1 - n_{kp}) \right)$
 $0 = \dot{n}_{pk} = \sum_p \left(\Gamma_{kp}^{\beta\beta} (n_{kp} - n_{pk}) + \Gamma_{kp}^{\beta\alpha} n_{kp} (1 - n_{pk}) - \Gamma_{pk}^{\alpha\beta} n_{pk} (1 - n_{kp}) \right)$
redundant?

*doleto
npru*

$(n_a, n_b, \Delta, m) \rightarrow \cdot n'_{ak} = n_{ak} + \delta t \cdot I_{ak}(n_a, n_b, \Delta, m)$

$\cdot 1 = \sum_k n'_{ak} + n'_{bk} \quad n'_{pk}$

$\cdot \Delta' = \sum_k \frac{V}{N} \frac{\Delta}{2\sqrt{\xi_k^2 + \Delta^2}} (n'_{bk} - n'_{ak})$

$\cdot m' = \sum_k (\mu_k^L - \nu_k^L) (n'_{ak} - n'_{bk})$

$\cdot \mu_k^L = \frac{1}{2} \left(1 + \frac{\xi_k}{\sqrt{\Delta'^2 + \xi_k^2}} \right) \quad , \quad \xi_k' = \frac{\bar{\xi}_a - \bar{\xi}_b}{2}$

detailed balance:

$$\frac{\Gamma_{i \leftarrow j}}{\Gamma_{j \leftarrow i}} = e^{-(E_i - E_j)/T} = e^{-(E_i - \mu)/T} e^{(\mu - E_j)/T} = \frac{f_i}{1 - f_i} \cdot \frac{1 - f_j}{f_j}$$

(FD): $\frac{f_i}{1 - f_i} = \frac{\frac{1}{1 + e^{\epsilon_i}}}{\frac{1}{1 + e^{\epsilon_j}}} = e^{-(\epsilon_i - \mu)/T}$

$$\rightarrow T_{j \leftarrow i} (1 - f_j) f_i = T_{i \leftarrow j} f_j (1 - f_i) \Rightarrow I = 0 \text{ (in total)}$$

python:
hamiltonian state

$$\Delta = V \Delta \sum_k w_k \frac{(n_{p_k} - n_{a_k})}{2 E_k}$$

$$w_k \propto \frac{1}{R}$$

$$E_k = \sqrt{\xi_k^2 + \Delta^2}$$

obs: $\Delta = 0 : E_k \approx |\xi_k| = \left| \frac{\epsilon_a - \epsilon_b}{2} \right|$
 $\xi_k = \epsilon_a + V n_a$

$$\Delta' \approx \Delta \sum_k w_k \frac{V}{2|\xi_k|} (n_{p_k} - n_{a_k}) = \Delta \lambda \quad \left\{ \begin{array}{ll} \lambda > 1: & \Delta' > \Delta \\ \lambda < 1: & \Delta' < \Delta \\ \lambda = 1: & \Delta' = \Delta \end{array} \right.$$

garni pshod!

$$\left. \begin{array}{l} E_{dk} = \eta_k + E_k \\ E_{pk} = \eta_k - E_k \end{array} \right\} - \frac{1}{2} = E_k \quad \checkmark$$

Bath ratios: $\Gamma_{k \leftarrow p}^{d/p} : \mu(p) \rightarrow \lambda(k)$

$$\Gamma_{k \leftarrow p} = \Gamma_{p \leftarrow k} = x \cdot e^{-\left(\frac{\Delta E}{w_k}\right)^2} \frac{1}{1 + e^{\Delta E/T}}$$

$$\Delta E = E_{dk} - E_{dp}$$

$$\frac{\Gamma_{k \leftarrow p}}{\Gamma_{p \leftarrow k}} = \frac{e^{-\left(\frac{E_k - E_{dp}}{w_k}\right)^2}}{e^{-\left(\frac{E_p - E_{dk}}{w_k}\right)^2}} \cdot \frac{\frac{1}{1 + e^{\frac{E_k - E_{dp}}{T}}}}{\frac{1}{1 + e^{\frac{E_p - E_{dk}}{T}}}} = \frac{\frac{1}{1 + e^x}}{\frac{e^x}{1 + e^x}} = e^{-x} \quad \checkmark$$

$$\Gamma_{k \leftarrow p} = e^{-\frac{E_k - E_{dp}}{T}} \Gamma_{p \leftarrow k}$$

→ intuitive: $T \downarrow$: transitions at high $E \rightarrow$ low E boy pshod

$T \uparrow$: pshodi iz $E \text{ high} \rightarrow E \text{ low}$ probitine evak

$$\frac{f_{xy}}{f_{yx}} = \frac{f\left(\frac{E_x}{T}\right) f\left(\frac{E_y}{T}\right)}{f\left(\frac{E_y}{T}\right) f\left(\frac{E_x}{T}\right)} = \frac{\frac{1}{1+e^{+x}} \cdot \frac{1}{1+e^{+y}}}{\frac{1}{1+e^{+y}} \cdot \frac{1}{1+e^{+x}}} = \frac{(1+e^{-x})(1+e^{-y})}{(1+e^{-y})(1+e^{-x})} =$$

$$= \frac{\cancel{e^{-x}}(1+\cancel{e^x})}{\cancel{e^{-y}}(1+\cancel{e^y})} = e^{-(x-y)} \checkmark$$